KARAN RAJENDRA
SOFTWARE ENGINEER(ML ENGINEER)

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EDUCATION

Stony Brook university Aug 2024 – May 2026
Master of Science in Computer Engineering — GPA: 3.9/4.0 *Stony Brook, NY*
Coursework: Machine Learning, Deep Learning, Natural Language Processing, Distributed Systems, Advanced Algorithms

Visvesvaraya Technological University Aug 2019 – Aug 2023
Bachelor of Engineering in Computer Science *Bangalore, India*

WORK EXPERIENCE

Research Assistant Jan 2024 – Present
Stony Brook University *Stony Brook, NY*

- Researched persuasion modeling by LoRA fine-tuning **Llama 2-7B** with DPO on **r/ChangeMyView**, achieving **0.7 reward accuracy**.
- Outperformed human responses by **6%** in a Qualtrics study through preference alignment on persuasive signals.
- Improved author-style detection performance from **87% to 92% ROC-AUC** using contrastive learning and Transformer-based models on **40M+ documents**.
- Increased speaker diarization accuracy from **46% to 96%** using **WhisperX** and **NVIDIA NeMo**, integrating multimodal ASR pipelines across **680+ audio interviews**.

Graduate Teaching Assistant Jan 2025 – May 2025
Stony Brook University *Stony Brook, NY*

- Mentored **40+ graduate students** in Machine Learning, providing guidance on model development, feature engineering, evaluation metrics, and debugging in Python.
- Conducted lab sessions on supervised learning, deep learning, and model evaluation techniques.
- Assisted students with ML projects involving data preprocessing, feature selection, model training, and performance evaluation.

Software Engineer Aug 2023 – June 2024
Meta *Bangalore, India*

- Engineered an end-to-end data pipeline and clustering framework for Instagram, processing **1.2B** user pairs to identify **500M** unique users and enabling the platform's first cohort analysis.
- Optimized large-scale data processing workflows by re-engineering pipeline logic, improving performance by **5x** while reducing Business Compute Unit (BCU) consumption by **50%**.
- Developed Instagram-specific behavioral signals, including story interactions, and implemented multi-signal clustering logic to address data sparsity and improve user segmentation.
- Designed and built scalable monitoring dashboards to track feature adoption and experiment performance, enabling data-driven product decisions across cross-functional teams.

PROJECTS

Watchlist Prediction (High-Risk Counterparties) Decision Trees, Predictive Modeling, Causal Inference, Statistics, Data Mining

- Prototyped a Random Forest with regression-based feature analysis and data mining on **100K+ records**, achieving **80% Gini**.
- Explored causal inference across **15+ client attributes** to identify key risk factors contributing to early watchlist inclusion.
- Simulated experimental evaluation with temporal splits and **5-fold backtesting**, projecting **$1.2B/year** in savings.

Speech Emotion Detection Application Deep Learning, FastAPI, Real-Time Inference, Google Cloud Run

- Developed a speech emotion classifier using a **CNN-BiLSTM-Attention** model deployed on GCP via Docker.
- Built a FastAPI backend and React frontend, styled with TailwindCSS and Framer Motion for an engaging user experience.
- Reduced API response latency by **40%** (**650ms → 390ms**) through asynchronous I/O, serving **100+** real-time inference requests.

SKILLS

Languages: Python, C++, Java, SQL, JavaScript, TypeScript, Scala
Machine Learning: PyTorch, TensorFlow, Scikit-learn, Hugging Face, NLP, Deep Learning, Transformers, LoRA
Data Science: NumPy, Pandas, SciPy, Statsmodels, Feature Engineering, Model Evaluation, Causal Inference
Cloud & MLOps: AWS, GCP, Azure, Docker, CI/CD, Airflow, TensorBoard, Weights & Biases
Backend & APIs: FastAPI, Flask, Node.js, REST APIs
Databases & Big Data: PostgreSQL, MySQL, MongoDB, Redis, Spark, Hive, HDFS
Visualization: Tableau, Matplotlib, Power BI